Lectures 26: Trace

MATH 110-3

Franny Dean

August 3/7, 2023

Definition

Recall:

Definition

Recall:

Def'n:

The **algebraic multiplicity** of an eigenvalue is the dimension of $G(\lambda, T)$.

Definition

Recall:

Def'n:

The **algebraic multiplicity** of an eigenvalue is the dimension of $G(\lambda, T)$.

Prop'n:

The sum of the multiplicities of the eigenvalues of T is the dimension of V.

Definition

Recall:

Def'n:

The **algebraic multiplicity** of an eigenvalue is the dimension of $G(\lambda, T)$.

Prop'n:

The sum of the multiplicities of the eigenvalues of T is the dimension of V.

Trace:

Suppose $T \in \mathcal{L}(V)$.

- If $\mathbb{F} = \mathbb{C}$, then the **trace** of T is the sum of the eigenvalues of T, with each eigenvalue repeated according to its multiplicity.

Definition

Recall:

Def'n:

The **algebraic multiplicity** of an eigenvalue is the dimension of $G(\lambda, T)$.

Prop'n:

The sum of the multiplicities of the eigenvalues of T is the dimension of V.

Trace:

Suppose $T \in \mathcal{L}(V)$.

- If $\mathbb{F} = \mathbb{C}$, then the **trace** of T is the sum of the eigenvalues of T, with each eigenvalue repeated according to its multiplicity.
- (If $\mathbb{F} = \mathbb{R}$, then the **trace** of T is the sum of the eigenvalues of the operator considered over $\mathbb{C}$ repeated according to multiplicity.)

Example

$T \in \mathcal{L}(\mathbb{C}^3)$ is the operator whose matrix is

$$\begin{pmatrix} 3 & -1 & -2 \\ 3 & 2 & -3 \\ 1 & 2 & 0 \end{pmatrix}.$$

Example

$T \in \mathcal{L}(\mathbb{C}^3)$ is the operator whose matrix is

$$\begin{pmatrix} 3 & -1 & -2 \\ 3 & 2 & -3 \\ 1 & 2 & 0 \end{pmatrix}.$$

The eigenvalues of T can be checked to be $1, 2+3i, 2-3i$ each with multiplicity 1.

Example

$T \in \mathcal{L}(\mathbb{C}^3)$ is the operator whose matrix is

$$\begin{pmatrix} 3 & -1 & -2 \\ 3 & 2 & -3 \\ 1 & 2 & 0 \end{pmatrix}.$$

The eigenvalues of T can be checked to be $1, 2+3i, 2-3i$ each with multiplicity 1.

So we have trace $T = 1 + 2 + 3i + 2 - 3i = 5$.

Example

$T \in \mathcal{L}(\mathbb{C}^3)$ is the operator whose matrix is

$$\begin{pmatrix} 3 & -1 & -2 \\ 3 & 2 & -3 \\ 1 & 2 & 0 \end{pmatrix}.$$

The eigenvalues of T can be checked to be $1, 2+3i, 2-3i$ each with multiplicity 1.

So we have trace $T = 1 + 2 + 3i + 2 - 3i = 5$.

What is the characteristic polynomial?

Example

$T \in \mathcal{L}(\mathbb{C}^3)$ is the operator whose matrix is

$$\begin{pmatrix} 3 & -1 & -2 \\ 3 & 2 & -3 \\ 1 & 2 & 0 \end{pmatrix}.$$

The eigenvalues of T can be checked to be $1, 2+3i, 2-3i$ each with multiplicity 1.

So we have trace $T = 1 + 2 + 3i + 2 - 3i = 5$.

What is the characteristic polynomial?

$$(z-1)(z-(2+3i))(z-(2-3i)) = z^3 - 5z^2 + 17z - 13$$

Trace and Characteristic Polynomial

Consider $T \in \mathcal{L}(V)$ for complex vector space V and eigenvalues $\lambda_1, \ldots, \lambda_n$.

Trace and Characteristic Polynomial

Consider $T \in \mathcal{L}(V)$ for complex vector space V and eigenvalues $\lambda_1, \ldots, \lambda_n$.

The characteristic polynomial is

$$(z - \lambda_1) \cdots (z - \lambda_n)$$

Trace and Characteristic Polynomial

Consider $T \in \mathcal{L}(V)$ for complex vector space V and eigenvalues $\lambda_1, \ldots, \lambda_n$.

The characteristic polynomial is

$$(z - \lambda_1) \cdots (z - \lambda_n)$$

which is

$$z^n - (\lambda_1 + \ldots + \lambda_n) z^{n-1} + \ldots + (-1)^n (\lambda_1 \cdots \lambda_n).$$

Trace and Characteristic Polynomial

Consider $T \in \mathcal{L}(V)$ for complex vector space V and eigenvalues $\lambda_1, \ldots, \lambda_n$.

The characteristic polynomial is

$$(z - \lambda_1) \cdots (z - \lambda_n)$$

which is

$$z^n - (\lambda_1 + \ldots + \lambda_n) z^{n-1} + \ldots + (-1)^n (\lambda_1 \cdots \lambda_n).$$

Prop'n:

Suppose $T \in \mathcal{L}(V)$. Let $n = \dim V$. Then trace T equals the negative of the coefficient of z^{n-1} in the characteristic polynomial of T.

Computing the Trace

- For upper triangular matrix form for an operator, the trace can be computed using the sum of the diagonal entries.

Computing the Trace

- For upper triangular matrix form for an operator, the trace can be computed using the sum of the diagonal entries.
- The same was true for our example even though not upper triangular...

Computing the Trace

- For upper triangular matrix form for an operator, the trace can be computed using the sum of the diagonal entries.
- The same was true for our example even though not upper triangular...

Def'n:

The **trace** of a square matrix A, denoted trace A, is defined to be the sum of the diagonal entries of A.

Computing the Trace

- For upper triangular matrix form for an operator, the trace can be computed using the sum of the diagonal entries.
- The same was true for our example even though not upper triangular...

Def'n:

The **trace** of a square matrix A, denoted trace A, is defined to be the sum of the diagonal entries of A.

Trace of an operator equals trace of its matrix:

Suppose $T \in \mathcal{L}(V)$. Then trace $T =$ trace $\mathcal{M}(T)$.

Computing the Trace

- For upper triangular matrix form for an operator, the trace can be computed using the sum of the diagonal entries.
- The same was true for our example even though not upper triangular...

Def'n:

The **trace** of a square matrix A, denoted trace A, is defined to be the sum of the diagonal entries of A.

Trace of an operator equals trace of its matrix:

Suppose $T \in \mathcal{L}(V)$. Then trace $T =$ trace $\mathcal{M}(T)$.

Let's show this!

Detour: Change of Basis

Let $I := \begin{pmatrix} 1 & & 0 \\ & \cdots & \\ 0 & & 1 \end{pmatrix}$

Detour: Change of Basis

Let $I := \begin{pmatrix} 1 & & 0 \\ & \cdots & \\ 0 & & 1 \end{pmatrix}$

Recall:

If $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$, then $\mathcal{M}(ST) = \mathcal{M}(S)\mathcal{M}(T)$.

Detour: Change of Basis

Let $I := \begin{pmatrix} 1 & & 0 \\ & \cdots & \\ 0 & & 1 \end{pmatrix}$

Recall:

If $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$, then $\mathcal{M}(ST) = \mathcal{M}(S)\mathcal{M}(T)$.

We can restate this with respect to bases as...

The matrix of the product of linear maps:

Suppose $u_1, \ldots, u_n$ and $v_1, \ldots, v_n$ and $w_1, \ldots, w_n$ are all bases of V. Suppose $S, T \in \mathcal{L}(V)$. Then

$$\mathcal{M}(ST, (u_1, \ldots, u_n), (w_1, \ldots, w_n)) =$$

$$\mathcal{M}(S, (v_1, \ldots, v_n), (w_1, \ldots, w_n))\mathcal{M}(T, (u_1, \ldots, u_n), (v_1, \ldots, v_n)).$$

Detour: Change of Basis

Thus, in our notation, we have

Matrix of the identity with respect to two bases:

Suppose $u_1, \ldots, u_n$ and $v_1, \ldots, v_n$ are bases of V. Then the matrices $\mathcal{M}(I, (u_1, \ldots, u_n), (v_1, \ldots, v_n))$ and $\mathcal{M}(I, (v_1, \ldots, v_n), (u_1, \ldots, u_n))$ are invertible and are inverses of each other.

Detour: Change of Basis

Thus, in our notation, we have

Matrix of the identity with respect to two bases:

Suppose $u_1, \ldots, u_n$ and $v_1, \ldots, v_n$ are bases of V. Then the matrices $\mathcal{M}(I, (u_1, \ldots, u_n), (v_1, \ldots, v_n))$ and $\mathcal{M}(I, (v_1, \ldots, v_n), (u_1, \ldots, u_n))$ are invertible and are inverses of each other.

Proof.

Detour: Change of Basis

Thus, in our notation, we have

Matrix of the identity with respect to two bases:

Suppose $u_1, \ldots, u_n$ and $v_1, \ldots, v_n$ are bases of V. Then the matrices $\mathcal{M}(I, (u_1, \ldots, u_n), (v_1, \ldots, v_n))$ and $\mathcal{M}(I, (v_1, \ldots, v_n), (u_1, \ldots, u_n))$ are invertible and are inverses of each other.

Proof. "Plug in" to previous result:

Detour: Change of Basis

Thus, in our notation, we have

Matrix of the identity with respect to two bases:

Suppose $u_1, \ldots, u_n$ and $v_1, \ldots, v_n$ are bases of V. Then the matrices $\mathcal{M}(I, (u_1, \ldots, u_n), (v_1, \ldots, v_n))$ and $\mathcal{M}(I, (v_1, \ldots, v_n), (u_1, \ldots, u_n))$ are invertible and are inverses of each other.

Proof. "Plug in" to previous result:

$$I = \mathcal{M}(I, (v_1, \ldots, v_n), (v_1, \ldots, v_n)) =$$

$$\mathcal{M}(I, (v_1, \ldots, v_n), (u_1, \ldots, u_n))\mathcal{M}(I, (u_1, \ldots, u_n), (v_1, \ldots, v_n))$$

Detour: Change of Basis

Thus, in our notation, we have

Matrix of the identity with respect to two bases:

Suppose $u_1, \ldots, u_n$ and $v_1, \ldots, v_n$ are bases of V. Then the matrices $\mathcal{M}(I, (u_1, \ldots, u_n), (v_1, \ldots, v_n))$ and $\mathcal{M}(I, (v_1, \ldots, v_n), (u_1, \ldots, u_n))$ are invertible and are inverses of each other.

Proof. "Plug in" to previous result:

$$I = \mathcal{M}(I, (v_1, \ldots, v_n), (v_1, \ldots, v_n)) =$$

$$\mathcal{M}(I, (v_1, \ldots, v_n), (u_1, \ldots, u_n))\mathcal{M}(I, (u_1, \ldots, u_n), (v_1, \ldots, v_n))$$

and

$$I = \mathcal{M}(I, (u_1, \ldots, u_n), (u_1, \ldots, u_n)) =$$

$$\mathcal{M}(I, (u_1, \ldots, u_n), (v_1, \ldots, v_n))\mathcal{M}(I, (v_1, \ldots, v_n), (u_1, \ldots, u_n))$$

Example

Consider the bases $(4, 2), (5, 3)$ and $(1, 0), (0, 1)$ of $\mathbb{F}^3$.

Example

Consider the bases $(4,2),(5,3)$ and $(1,0),(0,1)$ of $\mathbb{F}^3$.

$$\mathcal{M}(I,((4,2),(5,3)),((1,0),(0,1))) = \begin{pmatrix} 4 & 5 \\ 2 & 3 \end{pmatrix}$$

Example

Consider the bases $(4,2),(5,3)$ and $(1,0),(0,1)$ of $\mathbb{F}^3$.

$$\mathcal{M}(I,((4,2),(5,3)),((1,0),(0,1))) = \begin{pmatrix} 4 & 5 \\ 2 & 3 \end{pmatrix}$$

$$\mathcal{M}(I,((1,0),(0,1)),((4,2),(5,3))) = \begin{pmatrix} \frac{3}{2} & \frac{-5}{2} \\ -1 & 2 \end{pmatrix}$$

Change of Basis Formula

Prop'n:

Suppose $T \in \mathcal{L}(V)$. Let $u_1, \ldots, u_n$ and $v_1, \ldots, v_n$ be the bases of V. Let $A = \mathcal{M}(I, (u_1, \ldots, u_n), (v_1, \ldots, v_n))$. Then

$$\mathcal{M}(T, (u_1, \ldots, u_n)) = A^{-1}\mathcal{M}(T, (v_1, \ldots, v_n))A.$$

Change of Basis Formula

Prop'n:

Suppose $T \in \mathcal{L}(V)$. Let $u_1, \ldots, u_n$ and $v_1, \ldots, v_n$ be the bases of V. Let $A = \mathcal{M}(I, (u_1, \ldots, u_n), (v_1, \ldots, v_n))$. Then

$$\mathcal{M}(T, (u_1, \ldots, u_n)) = A^{-1}\mathcal{M}(T, (v_1, \ldots, v_n))A.$$

Proof.

Change of Basis Formula

Prop'n:

Suppose $T \in \mathcal{L}(V)$. Let $u_1, \ldots, u_n$ and $v_1, \ldots, v_n$ be the bases of V. Let $A = \mathcal{M}(I, (u_1, \ldots, u_n), (v_1, \ldots, v_n))$. Then

$$\mathcal{M}(T, (u_1, \ldots, u_n)) = A^{-1}\mathcal{M}(T, (v_1, \ldots, v_n))A.$$

Proof. Same idea.

Change of Basis Formula

Prop'n:

Suppose $T \in \mathcal{L}(V)$. Let $u_1, \ldots, u_n$ and $v_1, \ldots, v_n$ be the bases of V. Let $A = \mathcal{M}(I, (u_1, \ldots, u_n), (v_1, \ldots, v_n))$. Then

$$\mathcal{M}(T, (u_1, \ldots, u_n)) = A^{-1}\mathcal{M}(T, (v_1, \ldots, v_n))A.$$

Proof. Same idea.

$$\mathcal{M}(T, (u_1, \ldots, u_n)) = A^{-1}\mathcal{M}(T, (u_1, \ldots, u_n), (v_1, \ldots, v_n))$$

Change of Basis Formula

Prop'n:

Suppose $T \in \mathcal{L}(V)$. Let $u_1, \ldots, u_n$ and $v_1, \ldots, v_n$ be the bases of V. Let $A = \mathcal{M}(I, (u_1, \ldots, u_n), (v_1, \ldots, v_n))$. Then

$$\mathcal{M}(T, (u_1, \ldots, u_n)) = A^{-1}\mathcal{M}(T, (v_1, \ldots, v_n))A.$$

Proof. Same idea.

$$\mathcal{M}(T, (u_1, \ldots, u_n)) = A^{-1}\mathcal{M}(T, (u_1, \ldots, u_n), (v_1, \ldots, v_n))$$

and

$$\mathcal{M}(T, (u_1, \ldots, u_n), (v_1, \ldots, v_n)) = \mathcal{M}(T(v_1, \ldots, v_n))A$$

Change of Basis Formula

Prop'n:

Suppose $T \in \mathcal{L}(V)$. Let $u_1, \ldots, u_n$ and $v_1, \ldots, v_n$ be the bases of V. Let $A = \mathcal{M}(I, (u_1, \ldots, u_n), (v_1, \ldots, v_n))$. Then

$$\mathcal{M}(T, (u_1, \ldots, u_n)) = A^{-1}\mathcal{M}(T, (v_1, \ldots, v_n))A.$$

Proof. Same idea.

$$\mathcal{M}(T, (u_1, \ldots, u_n)) = A^{-1}\mathcal{M}(T, (u_1, \ldots, u_n), (v_1, \ldots, v_n))$$

and

$$\mathcal{M}(T, (u_1, \ldots, u_n), (v_1, \ldots, v_n)) = \mathcal{M}(T(v_1, \ldots, v_n))A$$

$\square$.

Back to Trace

Prop'n:

For A, B square matrices of the same size, trace $(AB) =$ trace (BA).

Back to Trace

Prop'n:

For A, B square matrices of the same size, trace $(AB) =$ trace (BA).

Proof.

Back to Trace

Prop'n:

For A, B square matrices of the same size, trace $(AB) =$ trace (BA).

Proof. Write $A = (A_{i,j})$ and $B = (B_{i,j})$.

Back to Trace

Prop'n:

For A, B square matrices of the same size, trace $(AB) =$ trace (BA).

Proof. Write $A = (A_{i,j})$ and $B = (B_{i,j})$.

The jth term on the diagonal of AB is $\sum_{k=1}^{n} A_{j,k} B_{k,j}$.

Back to Trace

Prop'n:

For A, B square matrices of the same size, trace $(AB) =$ trace (BA).

Proof. Write $A = (A_{i,j})$ and $B = (B_{i,j})$.

The jth term on the diagonal of AB is $\sum_{k=1}^{n} A_{j,k} B_{k,j}$.

So,

$$\text{trace } (AB) = \sum_{j=1}^{n} \sum_{k=1}^{n} A_{j,k} B_{k,j}$$

Back to Trace

Prop'n:

For A, B square matrices of the same size, trace $(AB) =$ trace (BA).

Proof. Write $A = (A_{i,j})$ and $B = (B_{i,j})$.

The jth term on the diagonal of AB is $\sum_{k=1}^{n} A_{j,k} B_{k,j}$.

So,

$$\begin{aligned} \text{trace } (AB) &= \sum_{j=1}^{n} \sum_{k=1}^{n} A_{j,k} B_{k,j} \\ &= \sum_{k=1}^{n} \sum_{j=1}^{n} B_{k,j} A_{j,k} \end{aligned}$$

Back to Trace

Prop'n:

For A, B square matrices of the same size, trace $(AB) =$ trace (BA).

Proof. Write $A = (A_{i,j})$ and $B = (B_{i,j})$.

The jth term on the diagonal of AB is $\sum_{k=1}^{n} A_{j,k}B_{k,j}$.

So,

$$\text{trace } (AB) = \sum_{j=1}^{n}\sum_{k=1}^{n} A_{j,k}B_{k,j}$$

$$= \sum_{k=1}^{n}\sum_{j=1}^{n} B_{k,j}A_{j,k}$$

$$= \sum_{k=1}^{n} k^{th} \text{ term on the diagonal of } AB = \text{trace } (BA)$$

Trace of a matrix of operator does not depend on basis

Prop'n:

Let $T \in \mathcal{L}(V)$. Suppose $u_1, \ldots, u_n$ and $v_1, \ldots, v_n$ are bases of V. Then

$$\operatorname{trace} \mathcal{M}(T, (u_1, \ldots, u_n)) = \operatorname{trace} \mathcal{M}(T, (v_1, \ldots, v_n)).$$

Trace of a matrix of operator does not depend on basis

Prop'n:

Let $T \in \mathcal{L}(V)$. Suppose $u_1, \ldots, u_n$ and $v_1, \ldots, v_n$ are bases of V. Then

$$\operatorname{trace} \mathcal{M}(T, (u_1, \ldots, u_n)) = \operatorname{trace} \mathcal{M}(T, (v_1, \ldots, v_n)).$$

Proof.

Trace of a matrix of operator does not depend on basis

Prop'n:

Let $T \in \mathcal{L}(V)$. Suppose $u_1, \ldots, u_n$ and $v_1, \ldots, v_n$ are bases of V. Then

$$\operatorname{trace} \mathcal{M}(T, (u_1, \ldots, u_n)) = \operatorname{trace} \mathcal{M}(T, (v_1, \ldots, v_n)).$$

Proof. Let $A = \mathcal{M}(I, (u_1, \ldots, u_n), (v_1, \ldots, v_n))$ again.

Trace of a matrix of operator does not depend on basis

Prop'n:

Let $T \in \mathcal{L}(V)$. Suppose $u_1, \ldots, u_n$ and $v_1, \ldots, v_n$ are bases of V. Then

$$\text{trace } \mathcal{M}(T, (u_1, \ldots, u_n)) = \text{trace } \mathcal{M}(T, (v_1, \ldots, v_n)).$$

Proof. Let $A = \mathcal{M}(I, (u_1, \ldots, u_n), (v_1, \ldots, v_n))$ again.

$$\begin{aligned}
\text{trace } \mathcal{M}(T, (u_1, \ldots, u_n)) &= \text{trace } (A^{-1}\mathcal{M}(T, (v_1, \ldots, v_n))A) \\
&= \text{trace } (\mathcal{M}(T, (v_1, \ldots, v_n))A^{-1}A) \\
&= \text{trace } (\mathcal{M}(T, (v_1, \ldots, v_n)))
\end{aligned}$$

Trace of a matrix of operator does not depend on basis

Prop'n:

Let $T \in \mathcal{L}(V)$. Suppose $u_1, \ldots, u_n$ and $v_1, \ldots, v_n$ are bases of V. Then

$$\operatorname{trace} \mathcal{M}(T, (u_1, \ldots, u_n)) = \operatorname{trace} \mathcal{M}(T, (v_1, \ldots, v_n)).$$

Proof. Let $A = \mathcal{M}(I, (u_1, \ldots, u_n), (v_1, \ldots, v_n))$ again.

$$\begin{aligned}
\operatorname{trace} \mathcal{M}(T, (u_1, \ldots, u_n)) &= \operatorname{trace} (A^{-1}\mathcal{M}(T, (v_1, \ldots, v_n))A) \\
&= \operatorname{trace} (\mathcal{M}(T, (v_1, \ldots, v_n))A^{-1}A) \\
&= \operatorname{trace} (\mathcal{M}(T, (v_1, \ldots, v_n)))
\end{aligned}$$

Corollary

Suppose $T \in \mathcal{L}(V)$. Then $\operatorname{trace} T = \operatorname{trace} \mathcal{M}(T)$.

Trace is a Linear Function(al)

Prop'n:

S'pose $S, T \in \mathcal{L}(V)$. Then trace $(S + T) =$ trace $S +$ trace T.

Trace is a Linear Function(al)

Prop'n:

S'pose $S, T \in \mathcal{L}(V)$. Then trace $(S + T) =$ trace $S +$ trace T.

Proof.

Trace is a Linear Function(al)

Prop'n:

S'pose $S, T \in \mathcal{L}(V)$. Then trace $(S + T) =$ trace $S +$ trace T.

Proof. Use the matrix definition!

Trace is a Linear Function(al)

Prop'n:

S'pose $S, T \in \mathcal{L}(V)$. Then trace $(S + T) =$ trace $S +$ trace T.

Proof. Use the matrix definition!

Prop'n:

S'pose $T \in \mathcal{L}(V)$ and $c \in \mathbb{F}$. Then trace $(cT) = c \cdot$ trace T.

Trace is a Linear Function(al)

Prop'n:

S'pose $S, T \in \mathcal{L}(V)$. Then trace $(S + T) =$ trace $S +$ trace T.

Proof. Use the matrix definition!

Prop'n:

S'pose $T \in \mathcal{L}(V)$ and $c \in \mathbb{F}$. Then trace $(cT) = c \cdot$ trace T.

Proof.

Trace is a Linear Function(al)

Prop'n:

S'pose $S, T \in \mathcal{L}(V)$. Then trace $(S + T) =$ trace $S +$ trace T.

Proof. Use the matrix definition!

Prop'n:

S'pose $T \in \mathcal{L}(V)$ and $c \in \mathbb{F}$. Then trace $(cT) = c \cdot$ trace T.

Proof. Use the matrix definition, again!

Trace is a Linear Function(al)

Prop'n:

S'pose $S, T \in \mathcal{L}(V)$. Then trace $(S + T) =$ trace $S +$ trace T.

Proof. Use the matrix definition!

Prop'n:

S'pose $T \in \mathcal{L}(V)$ and $c \in \mathbb{F}$. Then trace $(cT) = c \cdot$ trace T.

Proof. Use the matrix definition, again!

Application

Prop'n:

There do not exist operators S, T in $\mathcal{L}(V)$ such that $ST - TS = I$.

Application

Prop'n:

There do not exist operators S, T in $\mathcal{L}(V)$ such that $ST - TS = I$.

Proof.

Application

Prop'n:

There do not exist operators S, T in $\mathcal{L}(V)$ such that $ST - TS = I$.

Proof. S'pose $S, T \in \mathcal{L}(V)$.

Application

Prop'n:

There do not exist operators S, T in $\mathcal{L}(V)$ such that $ST - TS = I$.

Proof. S'pose $S, T \in \mathcal{L}(V)$. Choose a basis of V.

Application

Prop'n:

There do not exist operators S, T in $\mathcal{L}(V)$ such that $ST - TS = I$.

Proof. S'pose $S, T \in \mathcal{L}(V)$. Choose a basis of V. Then

$$\begin{aligned}
\text{trace}\,(ST - TS) &= \text{trace}\,(ST) - \text{trace}\,(TS) \\
&= \text{trace}\,\mathcal{M}(ST) - \text{trace}\,\mathcal{M}(TS) \\
&= \text{trace}\,\mathcal{M}(S)\mathcal{M}(T) - \text{trace}\,\mathcal{M}(T)\mathcal{M}(S) \\
&= 0
\end{aligned}$$

More Applications

Suppose $T \in \mathcal{L}(\mathbb{C}^3)$ with the matrix

$$\begin{pmatrix} 51 & -12 & -21 \\ 60 & -40 & -28 \\ 57 & -68 & 1 \end{pmatrix}.$$

More Applications

Suppose $T \in \mathcal{L}(\mathbb{C}^3)$ with the matrix

$$\begin{pmatrix} 51 & -12 & -21 \\ 60 & -40 & -28 \\ 57 & -68 & 1 \end{pmatrix}.$$

If someone tells you that -48 and 24 are eigenvalues of T, how could you tell what the third eigenvalue was?

References

[Axl14] Sheldon Axler.
Linear Algebra Done Right.
Undergraduate Texts in Mathematics. Springer Cham, 2014.